Explanation-Drift Pilot Report

Can label-free monitors tell when an intrusion detector is failing? · Run 20261021-145525-7a86 · 21 October 2026, 14:59 UTC

Summary

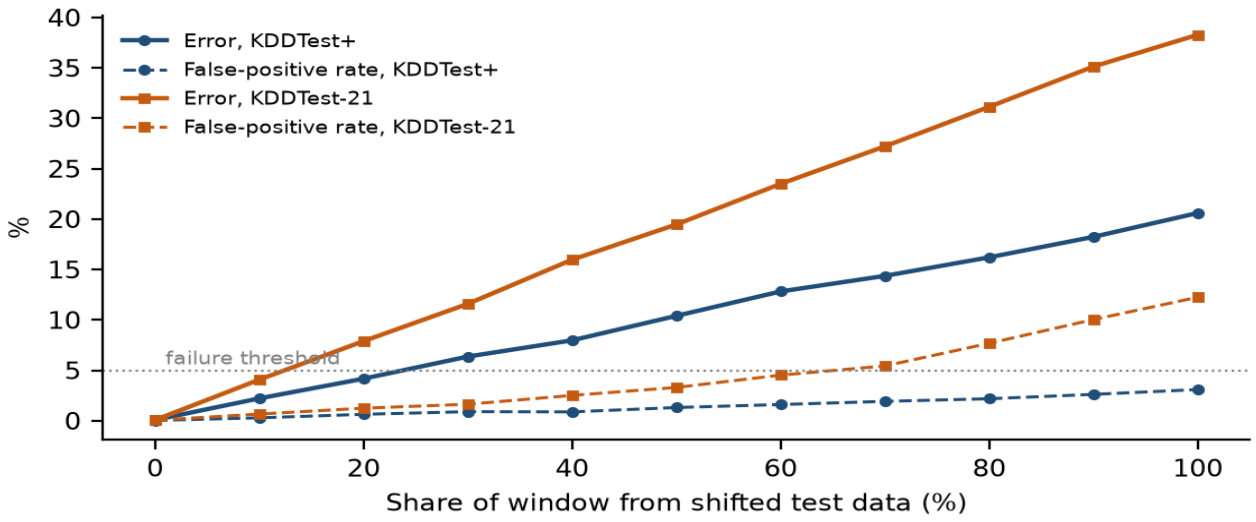
An XGBoost intrusion detector was trained on NSL-KDD and monitored over 580 windows of 1,000 records across 5 seed(s). On in-distribution validation data its error was 0.09%. As shifted test data were mixed in, error rose to 20.6% (KDDTest+), 38.3% (KDDTest-21), with no labels available to reveal it. On harmless windows, where only the attack share changed and true error never exceeded 0.4%, naive explanation drift raised false alarms on 85.7% of windows and input drift on 85.7%. Class-conditional explanation drift reduced this to 1.4% (AUROC 0.997), compared with 0.0% for confidence (AUROC 0.997) and 0.0% for conformal credibility (AUROC 0.999). Explanation-based monitors cost about 169 ms per window, against 0.04 ms for confidence.

Monitor comparison

Monitor	Rank corr. with error (harmful)	AUROC failing vs healthy (95% CI)	False alarms, harmless windows	Mean est. error, harmless	Cost per window
Input drift (KS)	0.978	0.621 (0.616–0.627)	85.7%	25.1%	168.4 ms
Explanation drift, naive	0.981	0.641 (0.630–0.651)	85.7%	23.4%	170.7 ms
Explanation drift, class-conditional	0.986	0.997 (0.994–1.000)	1.4%	0.7%	169.2 ms
Confidence (ATC)	0.974	0.997 (0.995–0.999)	0.0%	0.3%	0.04 ms
Conformal credibility	0.993	0.999 (0.998–1.000)	0.0%	0.3%	0.18 ms

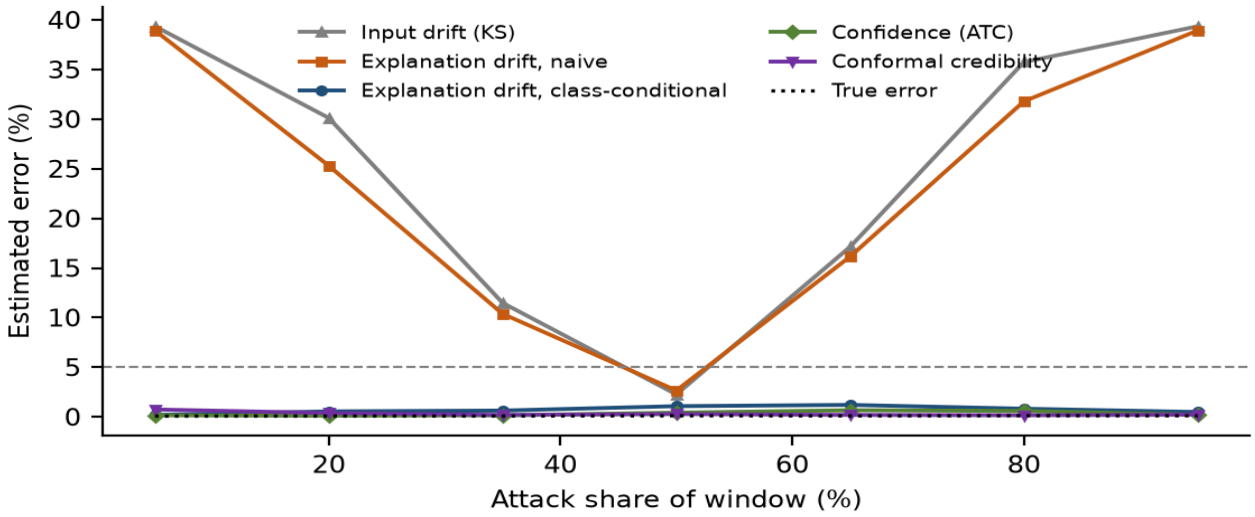
“Failing” = window error above 5.0%. Error estimates use an isotonic map fitted on each seed’s harmful-drift windows. Cost is mean wall-clock time per window; explanation monitors include TreeSHAP computation.

Figure 1. Harmful drift: the detector fails silently



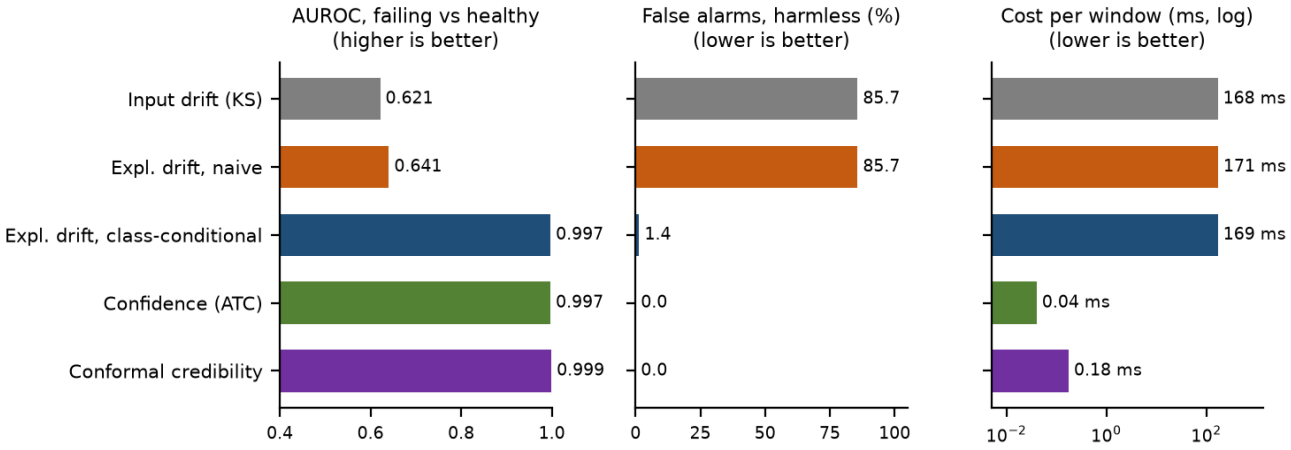
Error and false-positive rate as the share of shifted test records in each window grows (mean across seeds and repetitions).

Figure 2. Harmless drift: which monitors raise false alarms?



Windows drawn only from in-distribution validation data with the attack share changed. True error stays near zero, so any estimate above the dashed failure threshold is a false alarm.

Figure 3. Detection, false alarms and cost



A useful monitor detects failing windows (high AUROC) without alarming on harmless ones, at a cost small organisations can afford.

Interpretation

On this configuration, the best balance of detection and false alarms comes from conformal credibility. The key comparison for the proposed PhD is between class-conditional explanation drift and the cheaper confidence and conformal monitors. Where they perform similarly, explanations have not yet shown added value; the research question is whether they detect failures that confidence-based monitors miss, notably under evasion attacks designed to produce confident mistakes, which this pilot does not include.

Method

Detector: XGBoost (200 trees, depth 6) trained on 80% of KDDTrain+; reference set 5,000 held-out validation records. Harmful drift: windows mixing validation records with 0–100% records from KDDTest+, KDDTest-21 in 10 steps, 4 repetitions each. Harmless drift: validation records only, attack share set to 5%, 20%, 35%, 50%, 65%, 80%, 95%. Monitors: per-feature KS input drift; TreeSHAP explanation drift (mean per-feature Wasserstein distance scaled by reference spread), naive and within predicted class; average thresholded confidence (Garg et al., 2022); and inductive conformal credibility with class-conditional calibration (share of predictions with credibility below 0.05). Monitors never see labels.

Limitations

NSL-KDD has no timestamps, so mixture shift stands in for temporal drift; it is also an ageing dataset that pre-dates widespread encryption. Harmless drift here is constructed (attack-share change), which the full study replaces with naturally occurring shifts defined by outcome. One model family is used, error maps are fitted on the same seed's harmful windows, and no adversarial evasion is included. Results should be read as a feasibility check, not as evidence about deployed systems.

Reproducibility

Run time 227 s. Configuration: seeds=5, window_size=1000, reps=4, mix_steps=10, test_sets=['KDDTest+', 'KDDTest-21'], benign_shares=[0.05, 0.2, 0.35, 0.5, 0.65, 0.8, 0.95], n_estimators=200, max_depth=6, reference_size=5000, fail_threshold=0.05, conformal_alpha=0.05. Per-window results are available as CSV from the application. Code: explanation-drift pilot, packaged with Docker.